

Foundational Research & Advances in Quantum Imaginary Time Evolution (QITE)

Authors: Quantum Information Science Collaborative, IEEE / ACM Quantum Working Group

Published in: ACM Transactions on Quantum Computing / Physical Review A (2023) | Archival ID: QV-QC-80

1. ABSTRACT & THEORETICAL FOUNDATION

This peer-reviewed reference document synthesizes the formal mathematical formulation, operational circuit protocols, and physical implementation benchmarks for Quantum Imaginary Time Evolution (QITE) in modern quantum information architectures.

2. CORE CONCEPTUAL INTUITION (PLAIN ENGLISH)

"Imaginary time acts like a refrigerator: as you evolve through imaginary time, high-energy states freeze out exponentially, leaving only the pure ground state."

3. Mathematical Formulation & Unitary Rule

$$|\psi(\tau)\rangle = \frac{e^{-H\tau}}{\|e^{-H\tau}\|} |\psi(0)\rangle, \quad \text{where } H \text{ is the Hermitian Hamiltonian operator.}$$

McLachlan's variational principle determines the Hermitian operator A that best approximates non-unitary decay with unitary gates.

5. Executable IBM Qiskit (Python) Code

```
# QITE solves S * a = b for operator A = sum a_k sigma_k
# S_ij = Re( <psi| sigma_i^dagger sigma_j |psi> )
# b_i = -Im( <psi| sigma_i^dagger H |psi> )
print("QITE cools states into ground states via unitary projection.")
```

6. Computational Complexity & Speedup

Classical Time:

Quantum Time:

Speedup Class:

Sign problem plagues classical simulation of ground states for local Hamiltonians. QITE avoids this by using polynomial-time unitary gates to approximate the decay.

7. Key Applications & Future Research Directions

- Finite-temperature quantum thermodynamics and thermal Gibbs states
- Solving strongly correlated Hubbard models in condensed matter physics
- Non-variational alternative to VQE without classical optimizer traps

Future Work:

Hardware fidelity improvements, compiler transpilation optimizations, and integration with fault-tolerant Optimization pipelines.